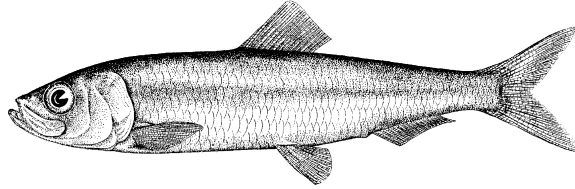


Pacific Herring preliminary data summary for Strait of Georgia 2025

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Pacific Herring (*Clupea pallasii*). Image credit: [Fisheries and Oceans Canada](#).

Disclaimer This report contains preliminary data collected for Pacific Herring in 2025 in the Strait of Georgia major stock assessment region (SAR). These data may differ from data used and presented in the final stock assessment.

1 Context

Pacific Herring (*Clupea pallasii*) in British Columbia are assessed as 5 major and 2 minor stock assessment regions (SARs), and data are collected and summarized on this scale (Table 1, Figure 1). Note that formal stock assessments are only done for major SARs. The Pacific Herring data collection program includes fishery-dependent and -independent data from 1951 to 2025. This includes annual time series of commercial catch data, biological samples (providing information on proportion-at-age and weight-at-age), and spawn index data conducted using a combination of surface and SCUBA surveys. In some areas, industry- and/or First Nations-operated in-season soundings programs are also conducted, and this information is used by resource managers, First Nations, and

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stakeholders to locate fish and identify areas of high and low Pacific Herring biomass to plan harvesting activities. In-season acoustic soundings are not used by stock assessment to inform the estimation of spawning biomass.

The following is a description of data collected for Pacific Herring in 2025 in the Strait of Georgia major SAR (Figure 2). Data collected outside the SAR boundary are not included in this summary, and are not used for the purposes of stock assessment. Although we summarise data at the scale of the SAR for stock assessments, we summarise data at finer spatial scales in this report: Locations are nested within Sections, Sections are nested within Statistical Areas, and Statistical Areas are nested within SARs (Table 2). For the Strait of Georgia major SAR, we use another level of spatial aggregation which we refer to as a ‘Group’. Note that we refer to ‘year’ instead of ‘herring season’ in this report; therefore 2025 refers to the 2024/2025 Pacific Herring season.

2 Data collection programs

Biological samples were collected by the seine charter vessel *Denman Isle* for 27 days from February 20th to March 18th. Four additional industry test vessels collected biological samples from March 1st to March 14th: *Viking Cavalier*, *Nita Maria*, *Western Investor*, and *Alaska Queen*. The primary purpose of the test charter vessel was to collect biological samples from main aggregations of herring in the SoG, as identified from soundings within the SAR boundary.

Herring spawn locations were primarily identified with fixed-wing overflights conducted by DFO Resource Management and Science staff. Fourteen flights were conducted this season, all in March.

Four dive vessels operated in the SoG:

- The *Pacific Discovery* surveyed 20 days from March 13th to April 2nd,
- The *Ocean Cloud* surveyed 12 days from March 13th to March 24th,
- Divers with A-Tlegay Fisheries Society surveyed 1 day on March 11th

The first two dive vessels and the seine charter vessel *Denman Isle* were funded by DFO, through a contract to the Herring Conservation Research Society. Additional sampling and sounding efforts conducted through the Industry Test Program were contributed in-kind by the Herring Industry.

3 Catch and biological samples

In the 1950s and 1960s, the reduction fishery dominated Pacific Herring catch; starting in the 1970s, catch has been predominantly from roe seine and gillnet fisheries. The reduction fishery is different from current fisheries in several ways. First, the reduction fishery caught Pacific Herring of all ages, whereas current fisheries target spawning (i.e., mature) fish. Thus, reduction fisheries included an unknown number of age-1 fish which

are not typically caught in current fisheries. Second, the reduction fishery has some uncertainty regarding the quantity and location of catch; however catches have been resolved to SAR and Statistical Area using fish slips as best as possible. For the roe gillnet fishery, all Pacific Herring catch has been validated by a dockside monitoring program since 1998; the catch validation program started in 1999 for the roe seine fishery. Finally, the reduction fishery operated during the summer and winter months, whereas roe fisheries typically target spawning fish between February and April.

Landed commercial catch of Pacific Herring by year and fishery is shown in Table 3 and Figure 3. Total harvested spawn-on-kelp (SOK) in 2025 in the Strait of Georgia major SAR is shown in Table 4; we also calculate the estimated spawning biomass associated with SOK harvest. See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. Incidental mortality from fisheries and aquaculture activities is shown in Figure 4.

In 2025, 87 Pacific Herring biological samples were collected and processed for the Strait of Georgia major SAR (Table 5, Table 6). The locations in which the biological samples were collected are presented in Figure 5. Figure 6 shows the distribution of fish weight by year and sample type. Included herein are biological summaries of observed proportion-, number-, weight-, and length-at-age (Figure 7, Table 7, and Figure 8, respectively). We also show the percent change in weight and length for age-3 and age-6 fish (Figure 9 & Figure 10, respectively). Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. Only representative biological samples are included, where ‘representative’ indicates whether the Pacific Herring sample in the set accurately reflects the larger Pacific Herring school.

3.1 Nearshore genetics project

DFO initiated a nearshore genetics project in 2022 to better understand the spatial and temporal structure of Pacific Herring stocks in British Columbia. We do this by collecting genetic and biological data from actively spawning Pacific Herring (*Clupea pallasii*). Then we collect biological data (e.g., length, weight, age, sex) and genetic data from the fish. Finally, we analyse the biological and genetic data to better understand Pacific Herring. Genetic samples were collected in the Strait of Georgia major SAR in 2023 and 2024 (Figure 11). Results from the nearshore genetics project will be presented in a separate document when they are available.

4 Spawn survey data

Pacific Herring spawn surveys were conducted at 55 individual locations in 2025 in the Strait of Georgia major SAR (Table 8, and Figure 12). A summary of spawn from the last decade (2015 to 2024) is shown in Figure 13. Figure 14 shows spawn start date by decade and Group. Spawn surveys are conducted to estimate the spawn length, width, number of egg layers, and substrate type, and these data are used to estimate the index of spawning biomass (i.e., the spawn index; Figure 15, Figure 16, Figure 17, Figure 18,

Table 9, and Figure 19). See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Therefore, these data do not represent model estimates of spawning biomass, and are considered the minimum observed spawning biomass derived from egg counts. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

Some Pacific Herring Sections contribute more than others to the total spawn index, and the percentage contributed by Section varies yearly (Figure 18b, Figure 20). For example, in 2025, Section 142 contributed the most to the spawn index (41%). As with Sections, some Groups contribute more than others to the total spawn index (Figure 18c, Figure 21). An animation shows the spawn index by spawn survey location from 1951 to 2025 (Figure 22).

5 First Nations observations

The following observations were contributed by representatives of First Nations communities. These observations provide context and perspectives to this data report. In some cases we make minor edits for clarity and brevity, but we do not change the intent or substance of responses.

5.1 Tla’amin

Tla’amin guardians did not find any spawn within the TFA this year but they did find some eggs on seaweed near Kitty Coleman. The only herring observed by the guardians were in deep water off the southern point of Savary Island (Mace Pt), however, it could have been another species as they were not able to catch any fish.

5.2 Homalko

We made three trips into Bute Inlet headwaters and located two small spot spawns. The spawns likely occurred late February; the observed spawns were not active and following an inspection of the eggs under a microscope I estimated the eggs to be 8 to 12 days old. This is much lighter coverage than I would have anticipated based on previous years. The rock weed band was diminished in width and density around the head of the inlet, possibly due in part to the effects of the Elliot Slide sediment plume that has persisted in the head of the inlet for about 1.5 years following the slide. We observed several rafts of diving birds on the first two visits and expected that the spawn was yet to occur. However, there were fewer rafts of diving birds in the inlet than expected and unlike previous years there were no Trumpeter Swans at the head of the inlet. No bodies of fish were observed on the sounder during our transits.

Unfortunately we were delayed in our start up awaiting a new patrol vessel for the Homalco guardians/watchmen. In the end we made all three trips in the *Discovery*

Huntress with a near shore inspection using the ship's tender. The Homalco guardians/watchmen had the opportunity to find some spawn and go through the procedure to measure and assess.

We had a good look at the foreshore with relatively ideal conditions at low water from the head of the inlet down and across the top of Bear Bay on both the East and West shores, these being the areas of historical use.

5.3 Qualicum

The main objective of our time on the water this year was the development of a water sampling program within QFN traditional territories. We started on March 5th and returned to Campbell River on March 22nd, with 65 water samples done at various locations. These samples were done between extreme winds which would produce a lot of mixing. We encountered more wind than usual this year with prolonged wind both South East and North West, with the usual South West between opposites. We observed very little back up spawn while transiting between spawns; at best we found small balls of fish tight on the bottom with no large schools that usually show in the vicinity of spawning activity. Spawns seemed to be in traditional locations but they seemed to be smaller in length and breadth and for shorter periods of time. However, many of the spawns we observed were intense. The difference in fish observed may be connected to the unusual weather conditions which changed quickly and frequently. We did not observe the usual large body of fish that usually shows outside in the deep water off Bowser shore. Generally herring scatter prior to a big wind, gathering and schooling again a day or two after the wind goes down. Another anomaly observed was two nearshore spawns starting at high water in shallows that would be close to dry at the next low water; traditionally a spawn will start at low water. Not as much deep water spawning as observed in 2024. The later, larger spawns around French Creek had a lot of fish, but gillnet catches were small compared to the amount of fish observed indicating there were a lot of small fish spawning. Compared to the last several years, I saw far fewer fish and active spawn this past season.

6 General observations

The following observations were reported by DFO Resource Management staff and DFO Science staff. These observations provide additional context to this data report.

- Greater spatial and temporal distribution of spawn than past few years. First spawn was on February 25th near Halfmoon Bay and the last spawn was on April 4th at Cape Mudge.
- Third reported spawn at Fisgard Light in the last four years.

7 Tables

Table 1. Pacific Herring stock assessment regions (SARs) in British Columbia.

Name	Code	Type
Haida Gwaii	HG	Major
Prince Rupert District	PRD	Major
Central Coast	CC	Major
Strait of Georgia	SoG	Major
West Coast of Vancouver Island	WCVI	Major
Area 27	A27	Minor
Area 2 West	A2W	Minor

Table 2. Statistical Areas, Sections, and Groups for Pacific Herring in the Strait of Georgia major stock assessment region (SAR). Legend: ‘14&17’ is Statistical Areas 14 and 17 (excluding Section 173); ‘ESoG’ is eastern Strait of Georgia; ‘Lazo’ is above Cape Lazo; and ‘SDodd’ is South of Dodd Narrows

Region	Statistical Area	Section	Group
Strait of Georgia	13	132	Lazo
Strait of Georgia	13	135	Lazo
Strait of Georgia	14	140	14&17
Strait of Georgia	14	141	Lazo
Strait of Georgia	14	142	14&17
Strait of Georgia	14	143	14&17
Strait of Georgia	15	150	ESoG
Strait of Georgia	15	151	ESoG
Strait of Georgia	15	152	ESoG
Strait of Georgia	16	160	ESoG
Strait of Georgia	16	161	ESoG
Strait of Georgia	16	162	ESoG
Strait of Georgia	16	163	ESoG
Strait of Georgia	16	164	ESoG
Strait of Georgia	16	165	ESoG
Strait of Georgia	17	170	14&17
Strait of Georgia	17	171	14&17
Strait of Georgia	17	172	14&17
Strait of Georgia	17	173	SDodd
Strait of Georgia	18	180	SDodd
Strait of Georgia	18	181	SDodd
Strait of Georgia	18	182	SDodd

<i>Table 2 continued</i>			
Region	Statistical Area	Section	Group
Strait of Georgia	19	190	SDodd
Strait of Georgia	19	191	SDodd
Strait of Georgia	19	192	SDodd
Strait of Georgia	19	193	SDodd
Strait of Georgia	28	280	ESoG
Strait of Georgia	29	291	ESoG
Strait of Georgia	29	292	ESoG

Table 3. Total landed commercial catch of Pacific Herring in metric tonnes (t) by gear type in 2025 in the Strait of Georgia major stock assessment region (SAR). Legend: ‘Other’ represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; ‘RoeSN’ represents the roe seine fishery; and ‘RoeGN’ represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Note: data may be withheld due to privacy concerns (WP).

Gear	Catch (t)
Other	2,407
RoeSN	2,723
RoeGN	4,349

Table 4. Total harvested Pacific Herring spawn-on-kelp (SOK) in pounds (lb), and the associated estimate of spawning biomass in metric tonnes (t) from 2015 to 2025 in the Strait of Georgia major stock assessment region (SAR). See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. Note: data may be withheld due to privacy concerns (WP).

Year	Harvest (lb)	Spawning biomass (t)
2015	0	0
2016	0	0
2017	0	0
2018	0	0
2019	0	0
2020	0	0
2021	0	0
2022	0	0
2023	0	0
2024	0	0
2025	0	0

Table 5. Number of Pacific Herring biological samples processed from 2015 to 2025 in the Strait of Georgia major stock assessment region (SAR). Each sample is approximately 100 fish. Note: Nearshore samples are not used in stock assessments.

Year	Number of samples			
	Commercial	Test	Nearshore	Total
2015	120	38	0	158
2016	123	33	5	161
2017	121	27	0	148
2018	116	28	0	144
2019	125	35	0	160
2020	106	27	0	133
2021	89	25	0	114
2022	50	18	2	70
2023	73	21	11	105
2024	68	24	9	101
2025	66	20	1	87

Table 6. Number and type of Pacific Herring biological samples processed in 2025 in the Strait of Georgia major stock assessment region (SAR). Each sample is approximately 100 fish.

Type	Gear	Use	Number of samples
Commercial	Gillnet	Roe fishery	34
Commercial	Seine	Bait fishery	3
Commercial	Seine	Food fishery	10
Commercial	Seine	Roe fishery	19
Test	Other	Nearshore	1
Test	Seine	Test fishery	20

Table 7. Observed proportion-at-age for Pacific Herring from 2015 to 2025 in the Strait of Georgia major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

Year	Proportion-at-age								
	2	3	4	5	6	7	8	9	10
2015	0.103	0.374	0.362	0.088	0.046	0.023	0.002	0.001	0.000
2016	0.153	0.267	0.334	0.178	0.040	0.016	0.009	0.001	0.001
2017	0.100	0.304	0.267	0.215	0.089	0.018	0.006	0.002	0.000
2018	0.077	0.418	0.247	0.144	0.082	0.026	0.004	0.002	0.000
2019	0.238	0.349	0.222	0.110	0.051	0.023	0.006	0.001	0.000
2020	0.124	0.529	0.208	0.081	0.035	0.016	0.004	0.002	0.000
2021	0.052	0.372	0.391	0.124	0.041	0.013	0.005	0.001	0.001
2022	0.240	0.282	0.238	0.176	0.048	0.010	0.004	0.001	0.001
2023	0.106	0.390	0.227	0.161	0.097	0.013	0.005	0.000	0.000
2024	0.079	0.375	0.327	0.114	0.071	0.027	0.005	0.002	0.000
2025	0.064	0.372	0.317	0.163	0.057	0.017	0.009	0.001	0.000

Table 8. Pacific Herring spawn survey locations, start date, and spawn index in metric tonnes (t) in 2025 in the Strait of Georgia major stock assessment region (SAR). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (NAs).

Statistical Area	Section	Location name	Start date	Spawn index (t)
13	135	Cape Mudge	April 04	NA
14	141	Kitty Coleman Beach	March 05	3,862
14	141	Kye Bay	March 02	2,395
14	141	Little Rvr	March 03	2,077
14	142	Boyle Pt	March 06	440
14	142	Cape Lazo	March 05	5,125
14	142	Collishaw Pt	March 07	1,732
14	142	Fillongley Park	March 05	3,079
14	142	Ford Cv	March 25	380
14	142	Gravelly Bay	March 06	1,342
14	142	Komas Bluff	March 10	13,579
14	142	Mapleguard Pt	March 13	496
14	142	Mud Bay (Baynes)	March 10	2,628
14	142	Repulse Pt	March 07	886
14	142	Seal Islets	March 10	2,420
14	142	Shingle Spit	March 25	256
14	142	Ship Pen	March 10	235
14	142	Tralee Pt	March 07	582
14	142	Whalebone Pt	March 03	5,875
14	142	Willemar Bluff	March 06	729
14	143	Big Qualicum Rvr	March 13	3,906
14	143	Bowser	March 07	3,198
14	143	Columbia Beach	March 08	3,041
14	143	Cottam Pt	March 12	768
14	143	Englishman Rvr	March 10	2,718
14	143	French Cr	March 08	8,012
14	143	Little Qualicum Rvr	March 10	120
14	143	Madrona Pt	March 11	1,997
14	143	Mistaken Is	March 13	744
14	143	Northwest Bay	March 11	525
14	143	Nuttal Bay	March 13	789
14	143	Parksville Bay	March 10	3,020
14	143	Qualicum Bay	March 07	185
14	143	Qualicum Beach	March 10	992
14	143	Rathtrevor Beach	March 10	849
15	151	Toba Inlt	February 27	133
16	163	McNaughton Pt	February 24	NA
17	172	Blunden Pt	March 18	930

Table 8 continued

Statistical Area	Section	Location name	Start date	Spawn index (t)
17	172	Hammond Bay	March 18	599
17	172	Icarus Pt	March 18	1,492
17	172	Lagoon Hd	March 18	211
17	172	Lantzville	March 18	963
17	172	Neck Pt	March 18	1,565
17	172	Sunrise Beach	March 18	1,656
17	173	Clam Bay	March 22	586
17	173	Dayman Is	March 22	142
17	173	Fraser Pt	March 22	5,300
17	173	Hudson Is	March 22	116
17	173	North Cv	March 28	140
17	173	Pilkey Pt	March 18	1,879
17	173	Preedy Hrbr	March 22	3,179
17	173	Telegraph Hrbr	March 22	5
17	173	Yellow Pt	April 05	NA
19	193	Esquimalt Lgn	March 15	NA
28	280	Fishermans Cv	April 15	NA

Table 9. Summary of Pacific Herring spawn survey data from 2015 to 2025 in the Strait of Georgia major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Units: metres (m), and metric tonnes (t).

Year	Total length (m)	Mean width (m)	Mean number of egg layers	Spawn index (t)
2015	166,750	114	1.3	104,481
2016	118,300	157	1.2	129,501
2017	130,440	162	0.9	81,063
2018	87,745	168	1.4	91,938
2019	82,865	139	1.1	63,037
2020	154,345	133	1.4	116,150
2021	114,950	154	1.1	70,938
2022	78,995	168	1.4	86,114
2023	102,550	142	1.4	74,506
2024	123,325	129	1.4	81,774
2025	138,375	130	1.6	97,903

8 Figures

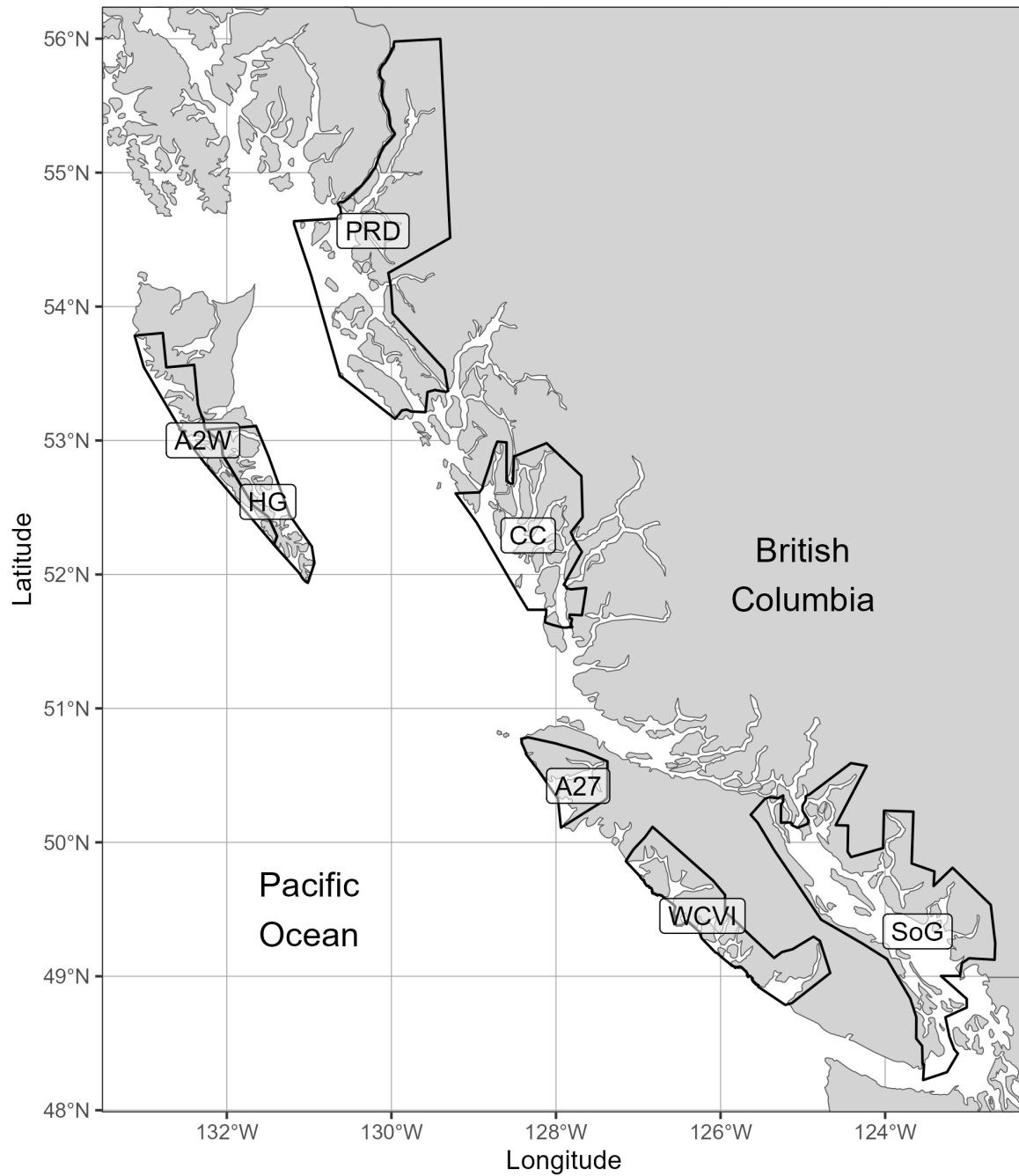


Figure 1. Boundaries for the Pacific Herring stock assessment regions (SARs) in British Columbia. There are 5 major SARs: Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), Strait of Georgia (SoG), and West Coast of Vancouver Island (WCVI). There are 2 minor SARs: Area 27 (A27) and Area 2 West (A2W). Units: kilometres (km).

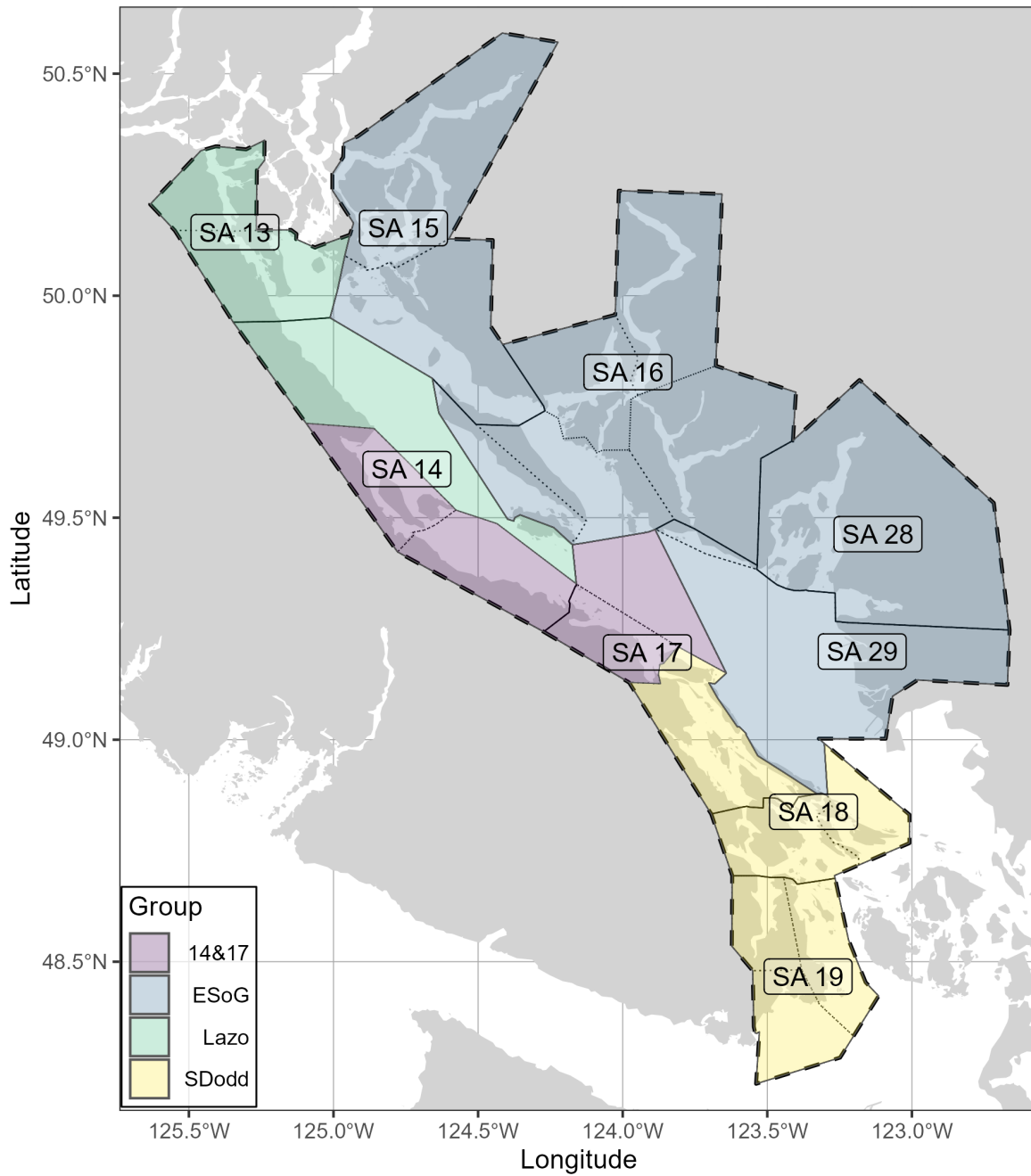


Figure 2. Boundaries for the Strait of Georgia major stock assessment region (SAR; thick dashed lines), associated Statistical Areas (SA; thin solid lines), and associated Sections (thin dotted lines). Units: kilometres (km). Legend: ‘14&17’ is Statistical Areas 14 and 17 (excluding Section 173); ‘ESoG’ is eastern Strait of Georgia; ‘Lazo’ is above Cape Lazo; and ‘SDodd’ is South of Dodd Narrows

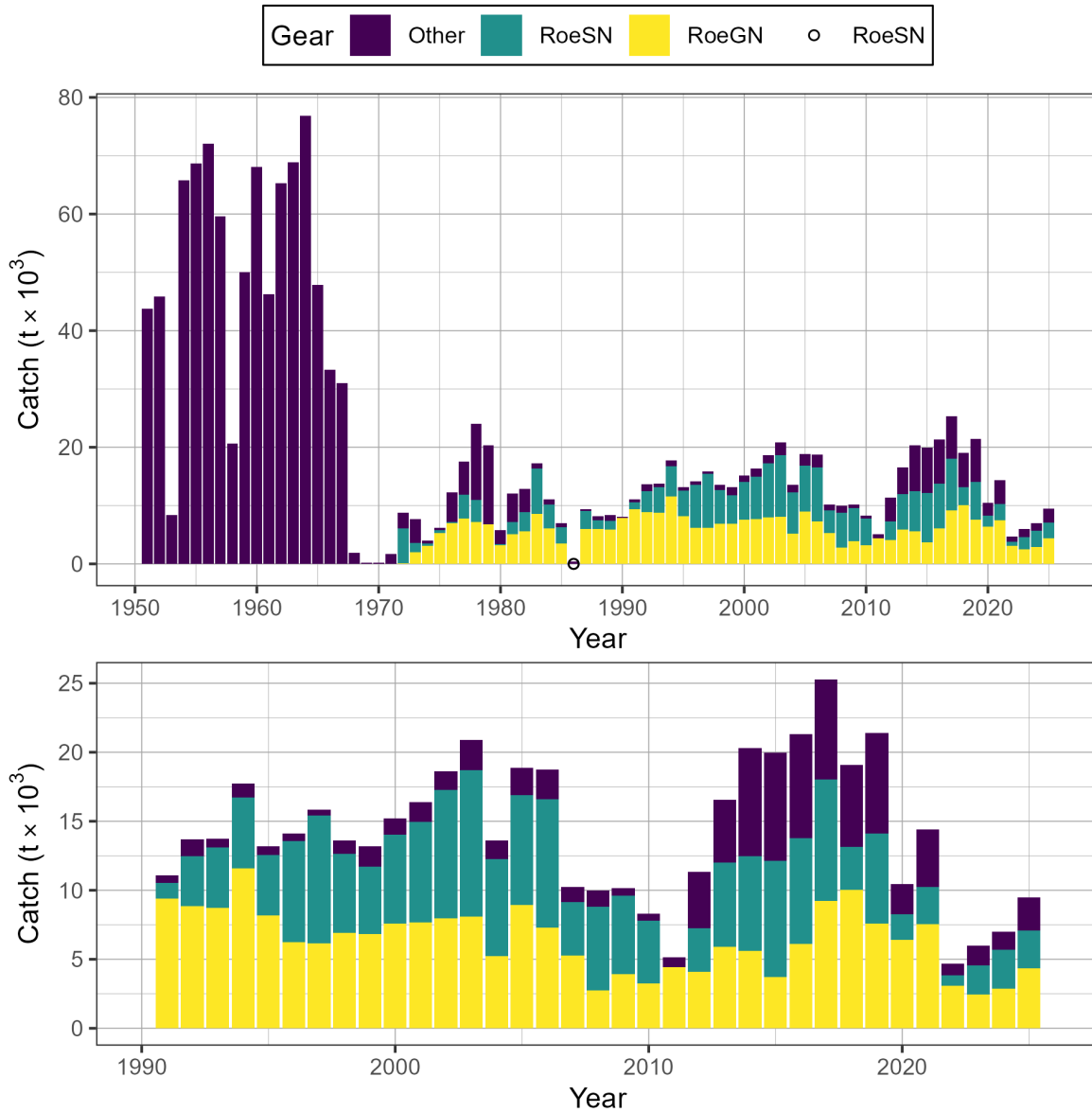


Figure 3. Time series of total landed catch in thousands of metric tonnes ($t \times 10^3$) of Pacific Herring by gear type from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). Legend: 'Other' represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; 'RoeSN' represents the roe seine fishery; and 'RoeGN' represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Bottom panel shows catch since 1990 in more detail. Note: symbols indicate years in which catch by gear type (i.e., Other, RoeSN, RoeGN) is withheld due to privacy concerns.

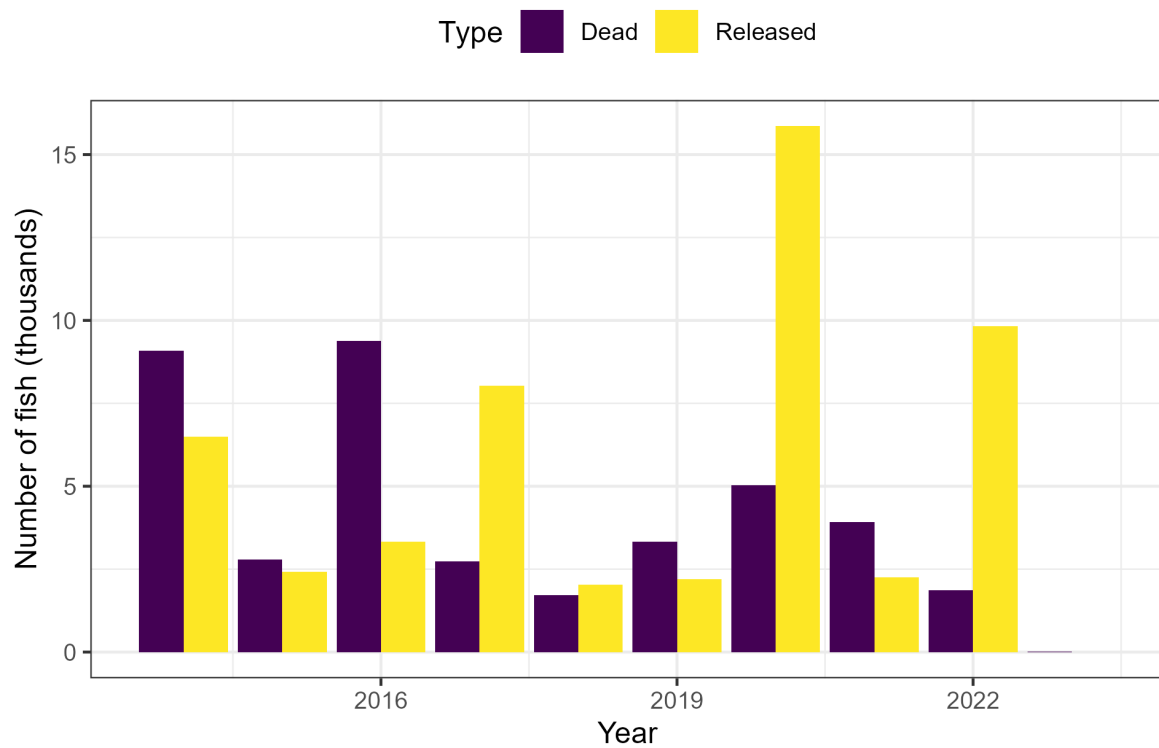


Figure 4. Incidental Pacific Herring mortality in aquaculture activities in thousands of fish from 2014 to 2025 in the Strait of Georgia major stock assessment region (SAR). Note: figure may include data outside SAR boundaries.

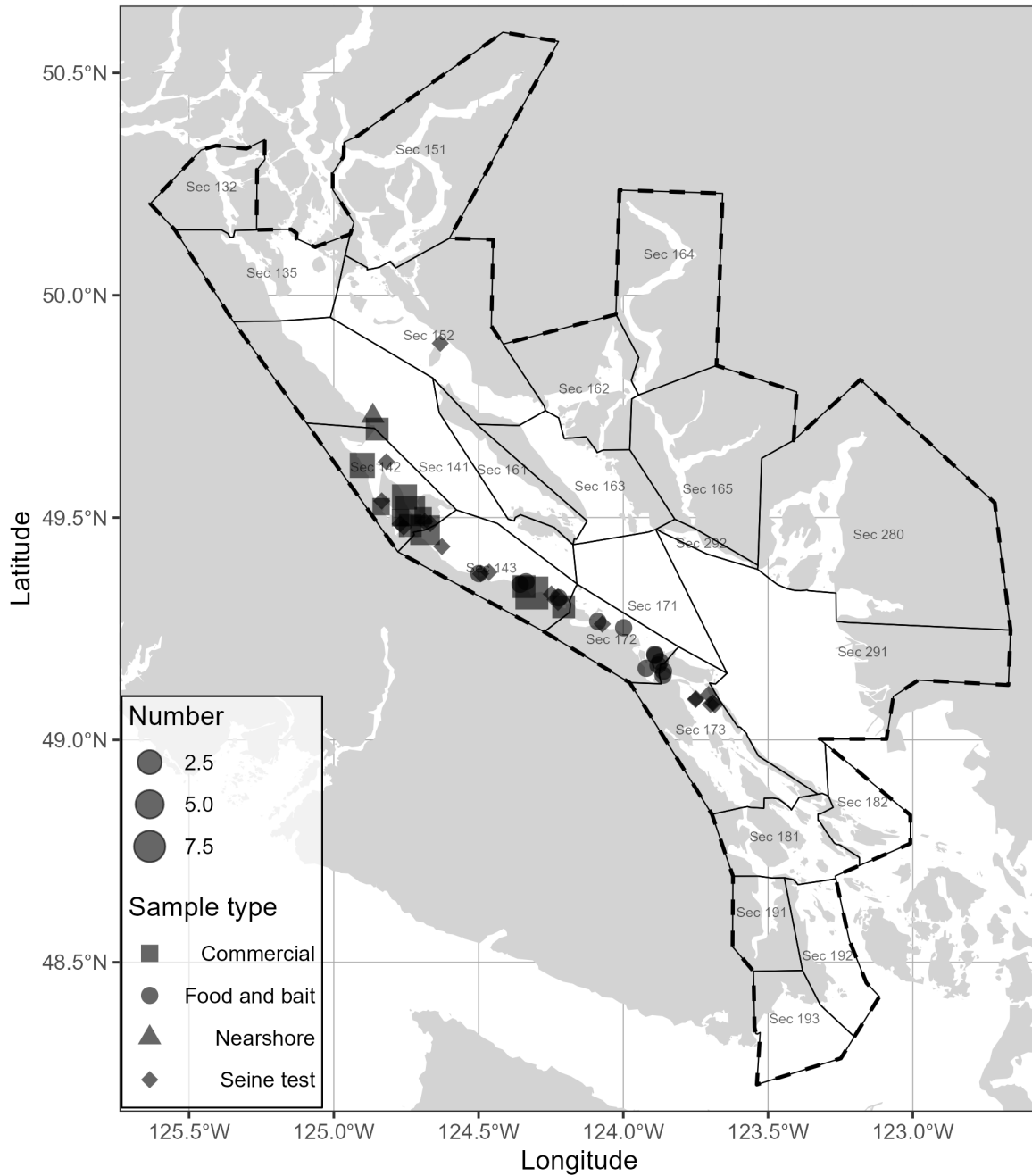


Figure 5. Location and type of Pacific Herring biological samples collected in 2025 in the Strait of Georgia major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). Units: kilometres (km).

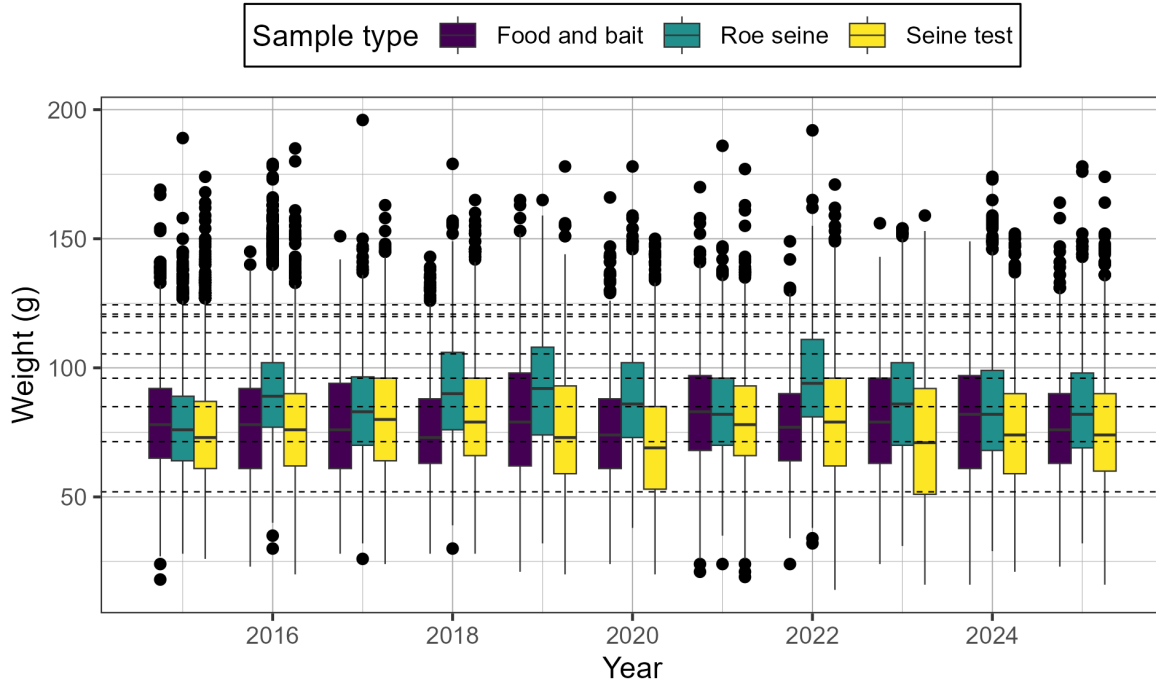


Figure 6. Time series of weight in grams (g) of Pacific Herring by sample type from 2015 to 2025 in the Strait of Georgia major stock assessment region (SAR) in Statistical Areas 14 and 17. The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to $1.5 \times \text{IQR}$, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Horizontal dashed lines indicate the mean weight-at-age for age-2 (lowest line) to age-10 (incrementing higher from age-2) fish. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

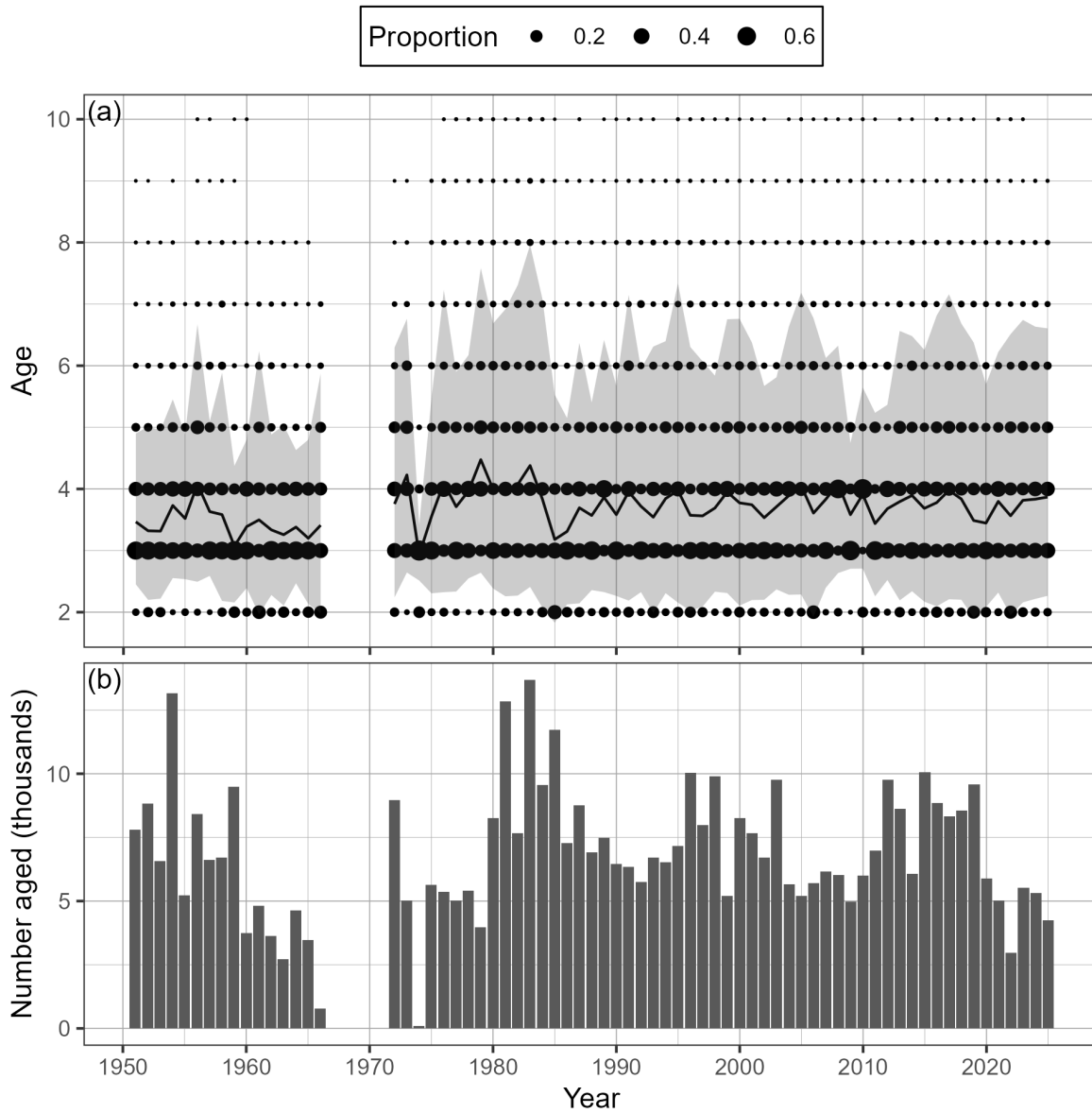


Figure 7. Time series of observed proportion-at-age (a) and number aged in thousands (b) of Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). The black line is the mean age, and the shaded area is the approximate 90% distribution. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

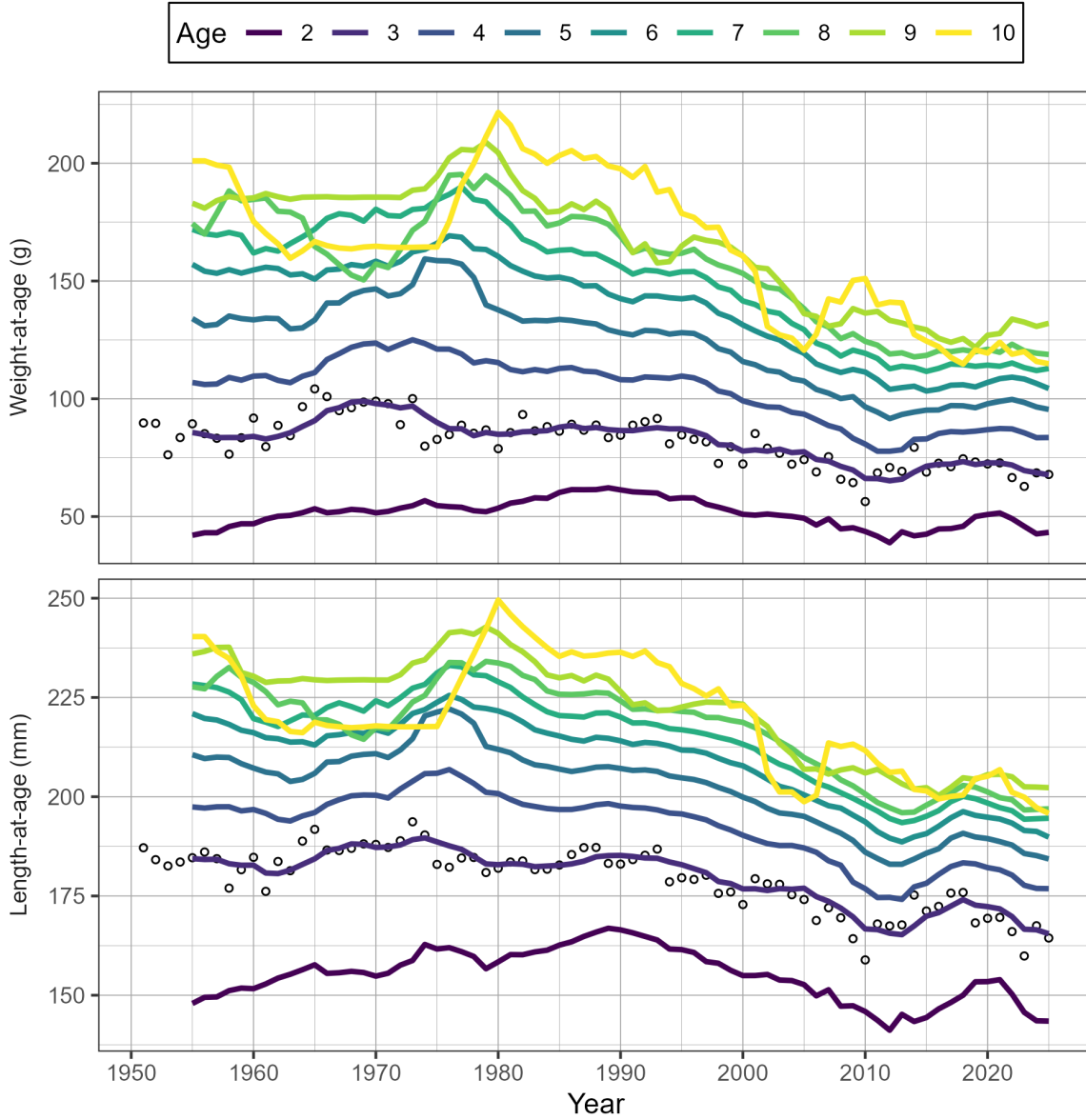


Figure 8. Time series of weight-at-age in grams (g) and length-at-age in millimetres (mm) for age-3 (circles) and 5-year running mean weight- and length-at-age (lines) for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). Missing weight- and length-at-age values (i.e., years with no biological samples) are imputed using one of two methods: missing values at the beginning of the time series are imputed by extending the first non-missing value backwards; other missing values are imputed as the mean of the previous 5 years. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

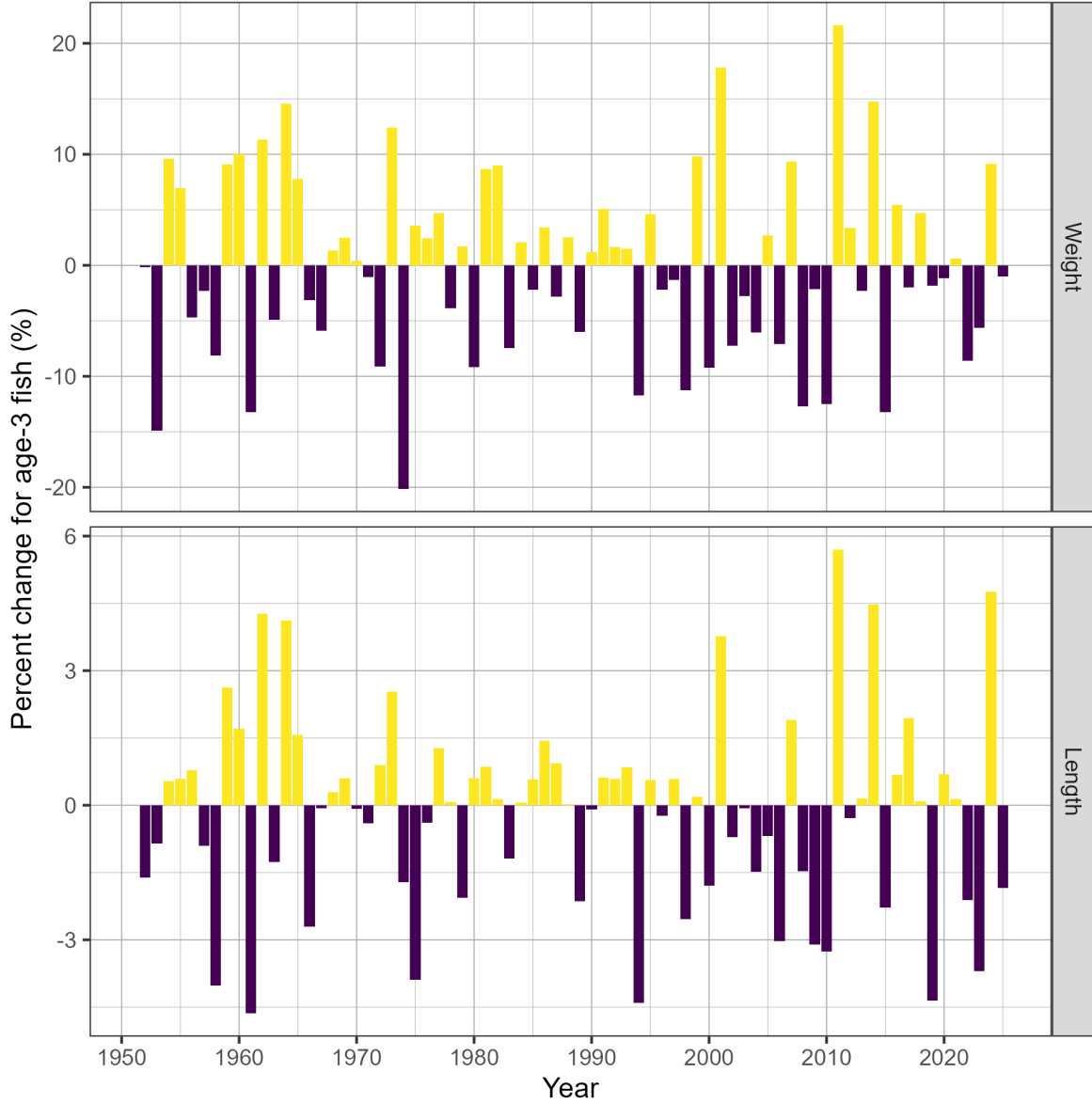


Figure 9. Time series of percent change (%) in weight and length for age-3 fish for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-3 fish, respectively, in year t . Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

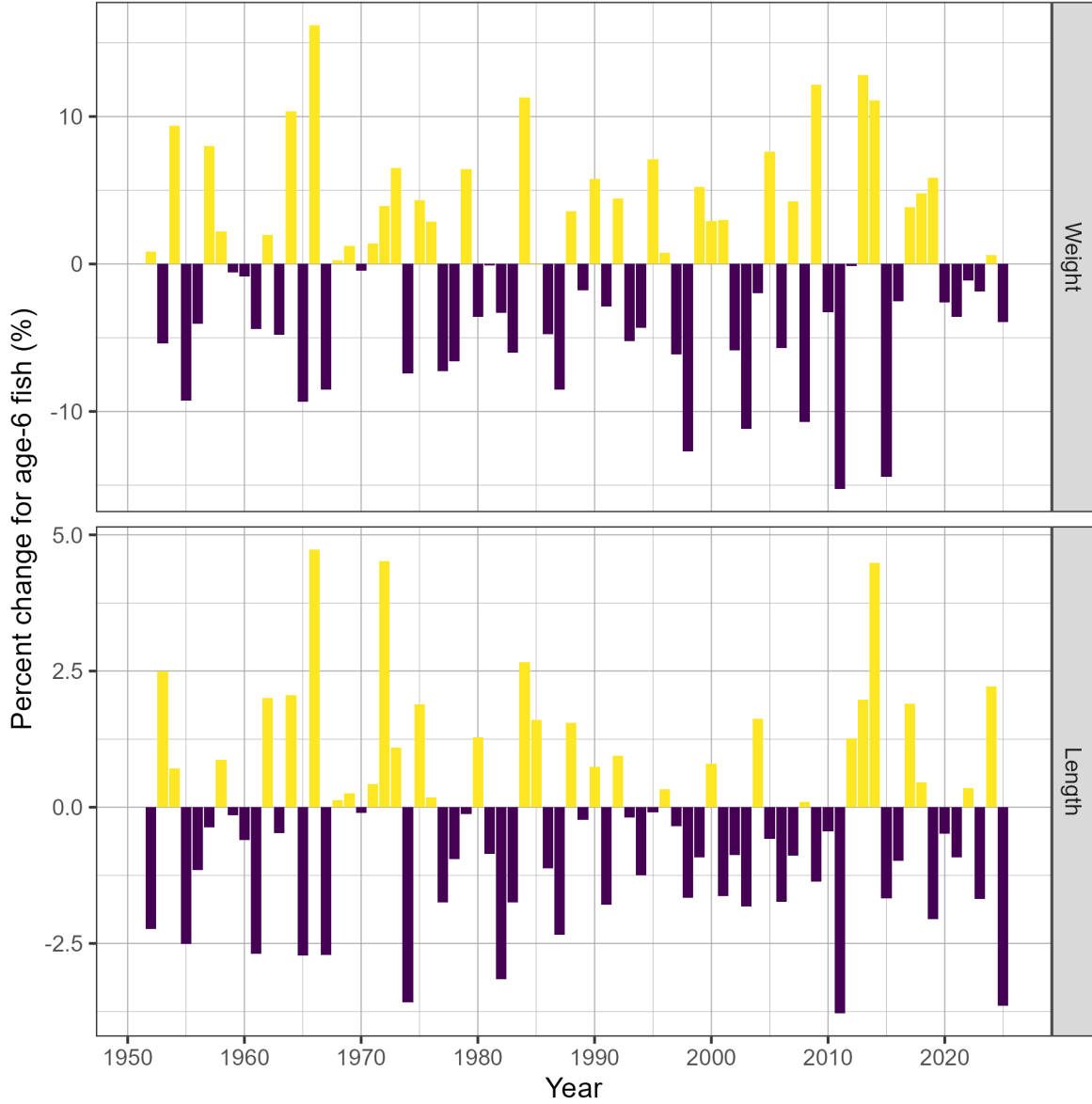


Figure 10. Time series of percent change (%) in weight and length for age-6 fish for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-6 fish, respectively, in year t . Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

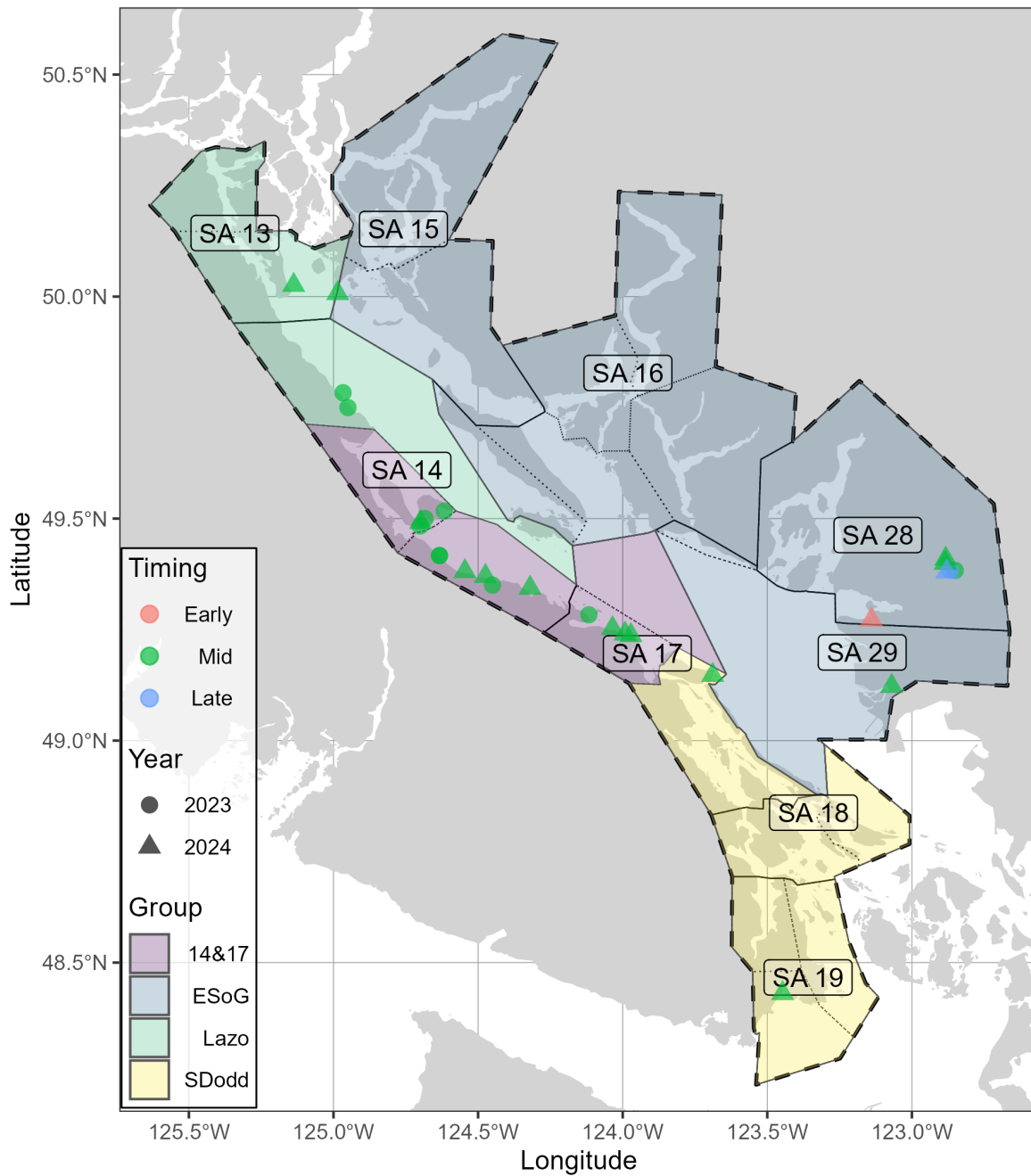


Figure 11. Nearshore genetics sampling in the Strait of Georgia major SAR. Note that timing indicates the collection date: ‘Early’ is January and February, ‘Mid’ is March and April, and ‘Late’ is May, June, and July. In addition, note that genetic samples from 2025 are not yet available. See Figure 2 for legend.

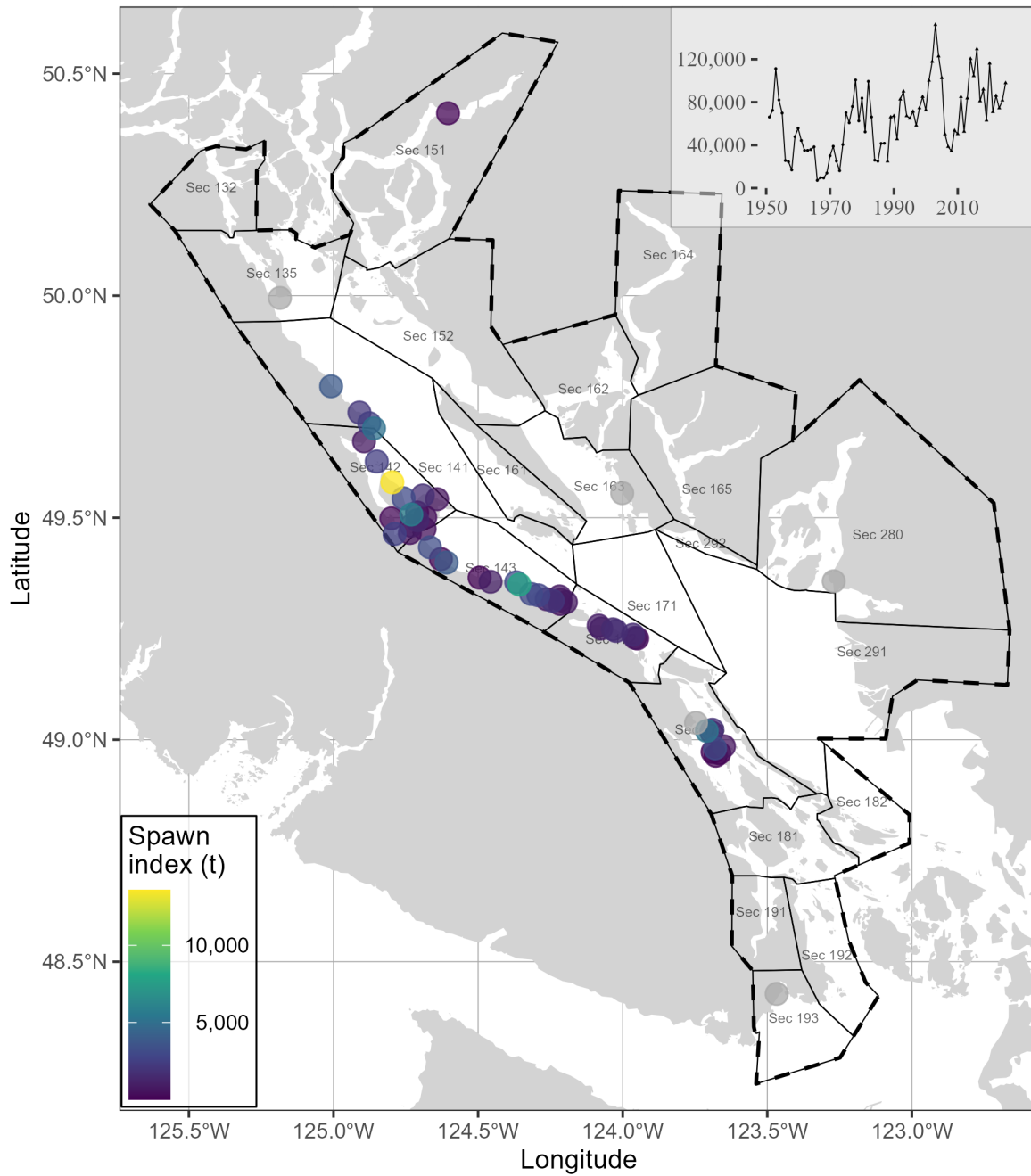


Figure 12. Pacific Herring spawn survey locations, and spawn index in metric tonnes (t) in 2025 in the Strait of Georgia major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km).

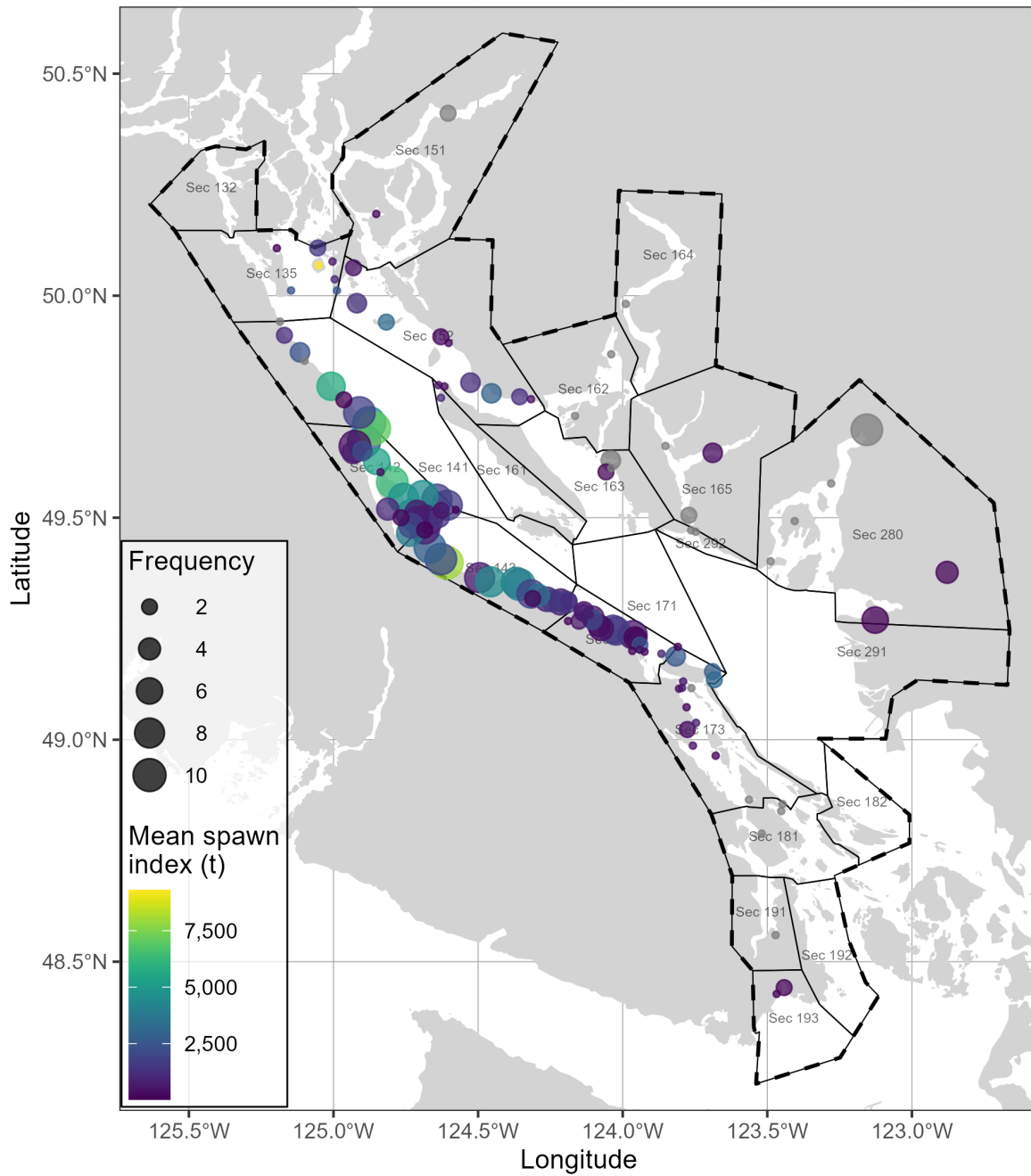


Figure 13. Pacific Herring spawn survey locations, mean spawn index in metric tonnes (t), and spawn frequency from 2015 to 2024 in the Strait of Georgia major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Units: kilometres (km).

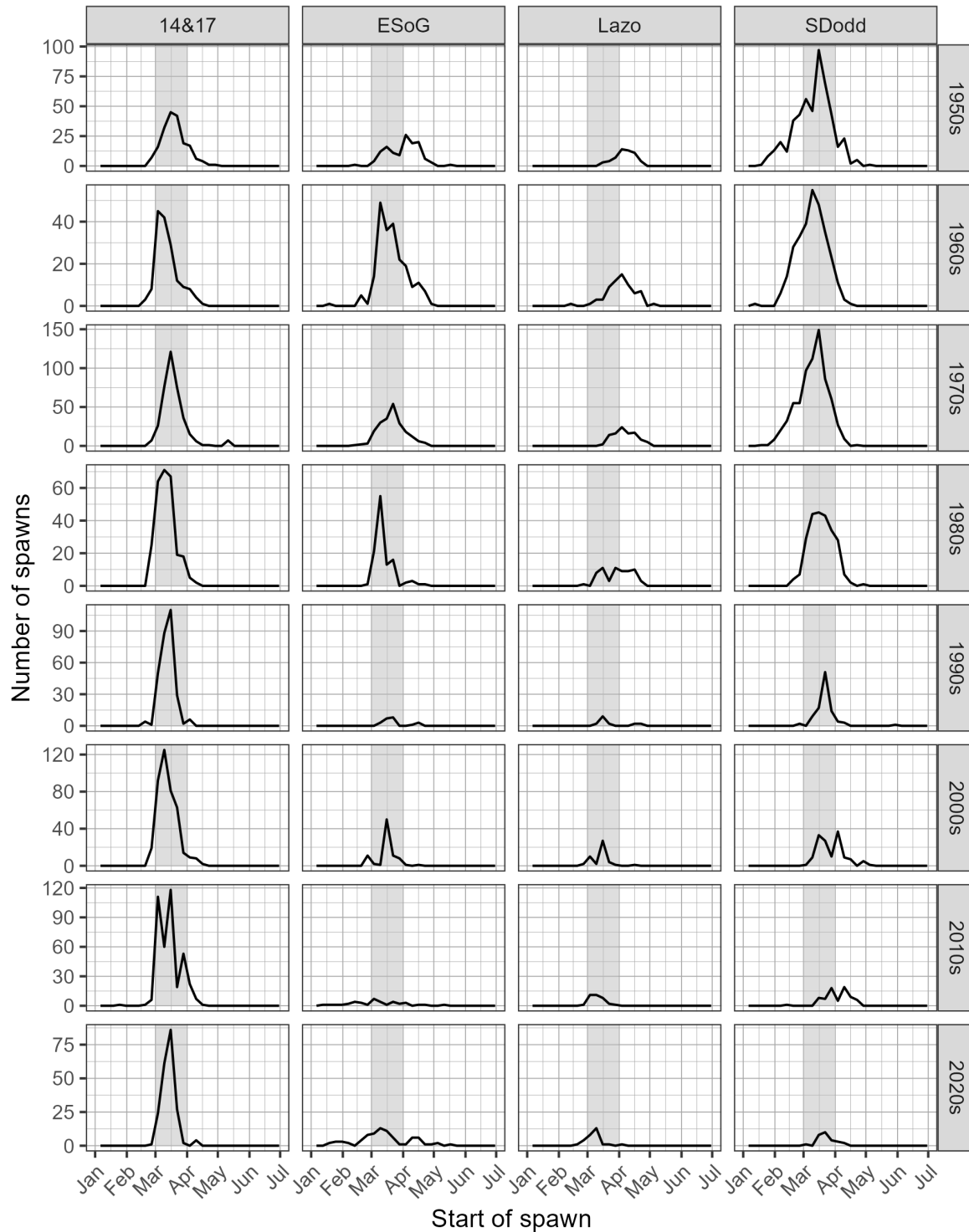


Figure 14. Pacific Herring spawn start date by decade and Group. Grey shaded regions indicate March 1st to 31st. Note that spawn size and intensity varies; therefore the number of spawns is not directly proportional to spawn extent or biomass. Legend: '14&17' is Statistical Areas 14 and 17 (excluding Section 173); 'ESoG' is eastern Strait of Georgia; 'Lazo' is above Cape Lazo; and 'SDodd' is South of Dodd Narrows

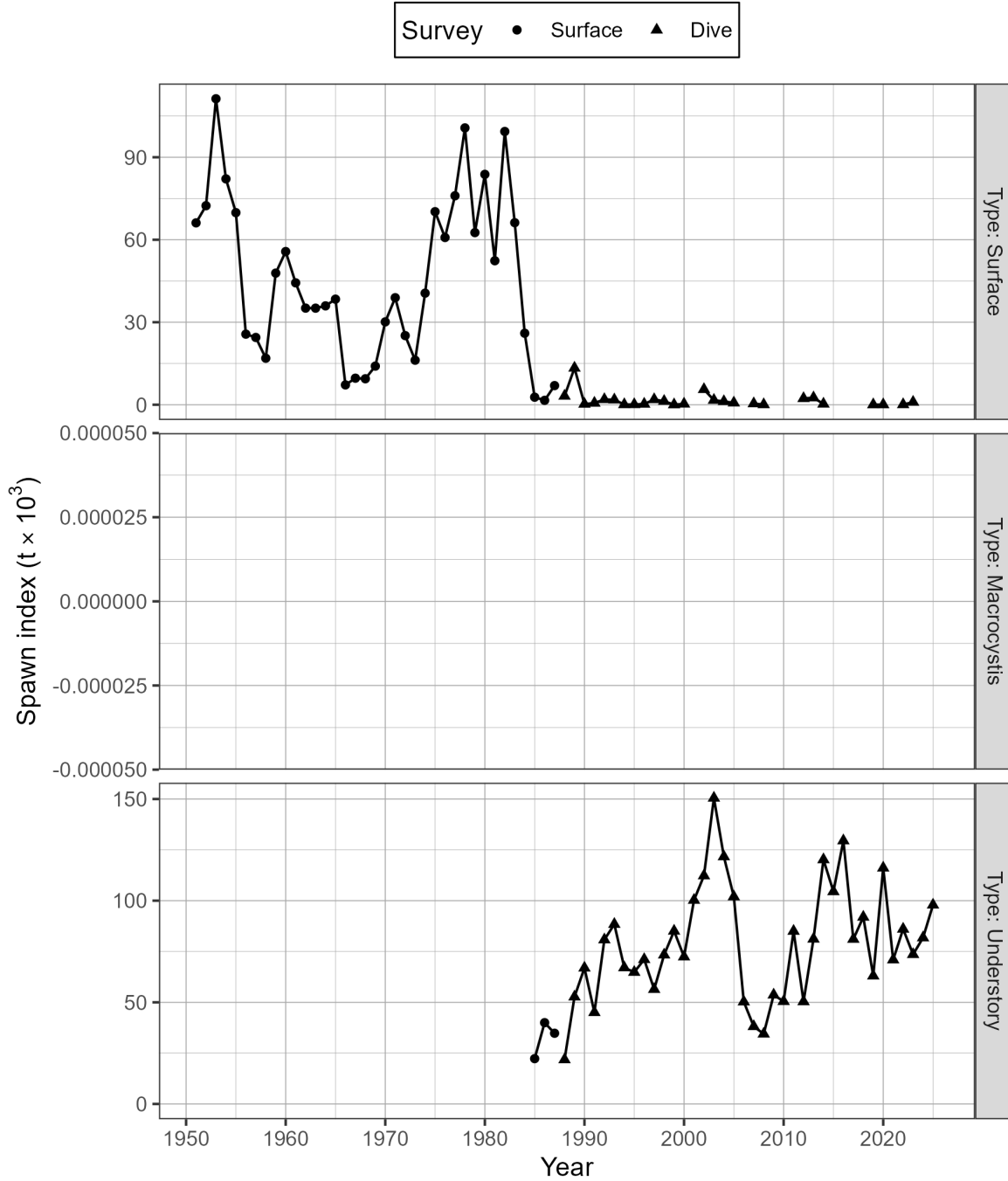


Figure 15. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) by type for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). There are three types of spawn survey observations: observations of spawn taken from the surface usually at low tide, underwater observations of spawn on giant kelp, *Macrocystis* (*Macrocystis* spp.), and underwater observations of spawn on other types of algae and the substrate, which we refer to as ‘understory.’ The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

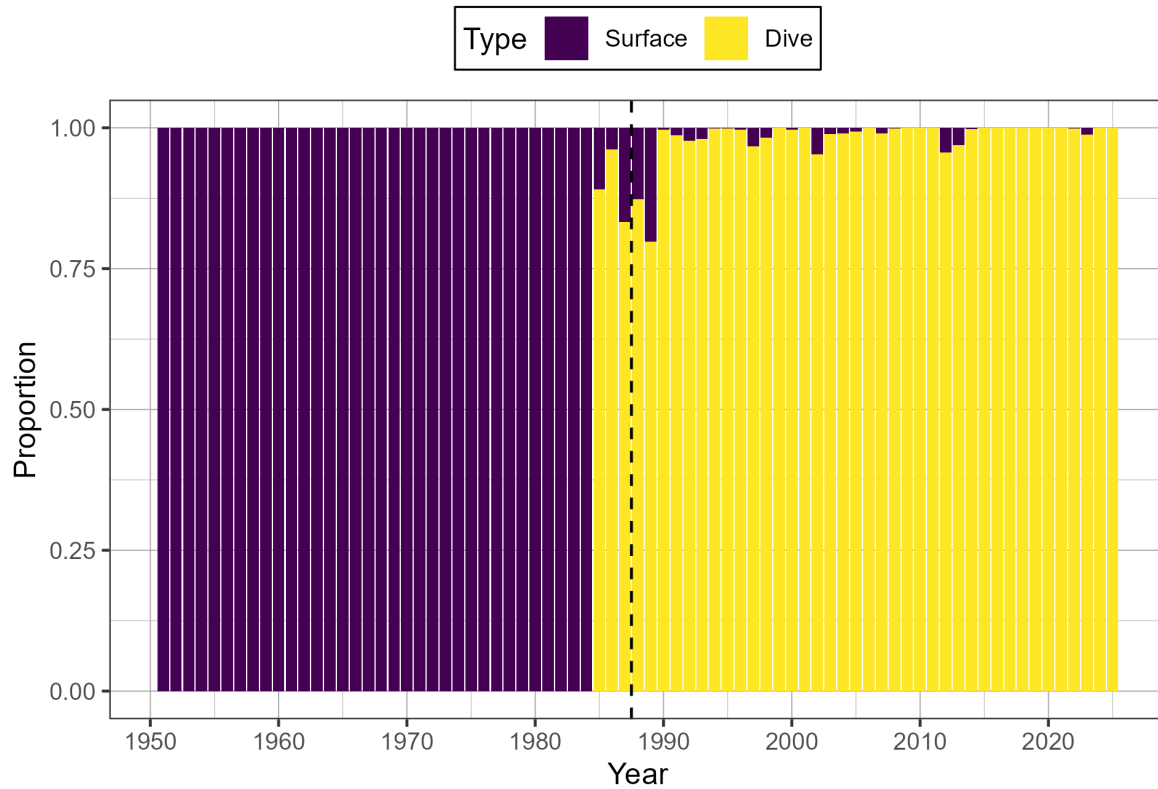


Figure 16. Time series of proportion of spawn index by surface and dive surveys for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

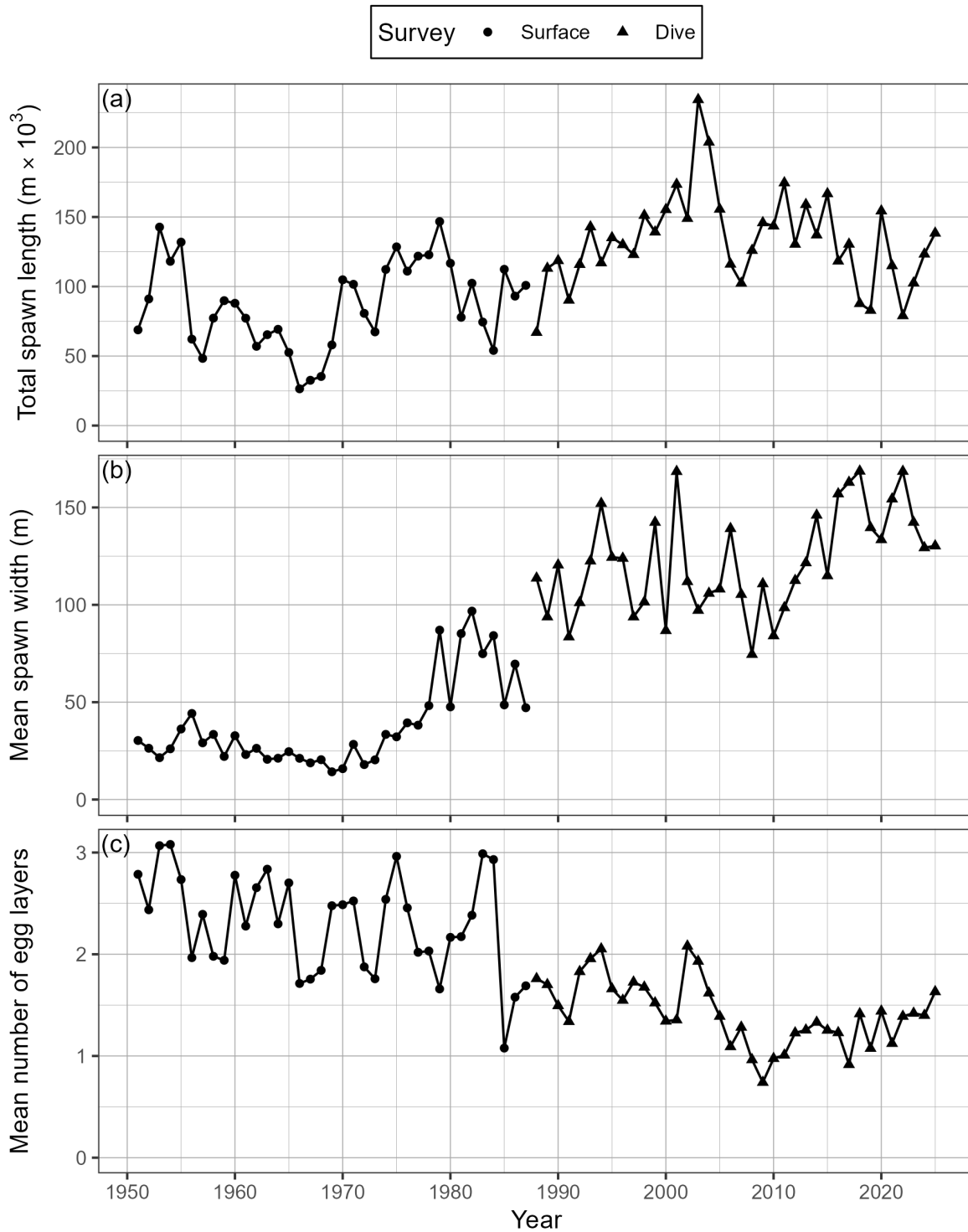


Figure 17. Time series of total spawn length in thousands of metres ($m \times 10^3$; panel a), mean spawn width in metres (b), and mean number of egg layers (c) for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

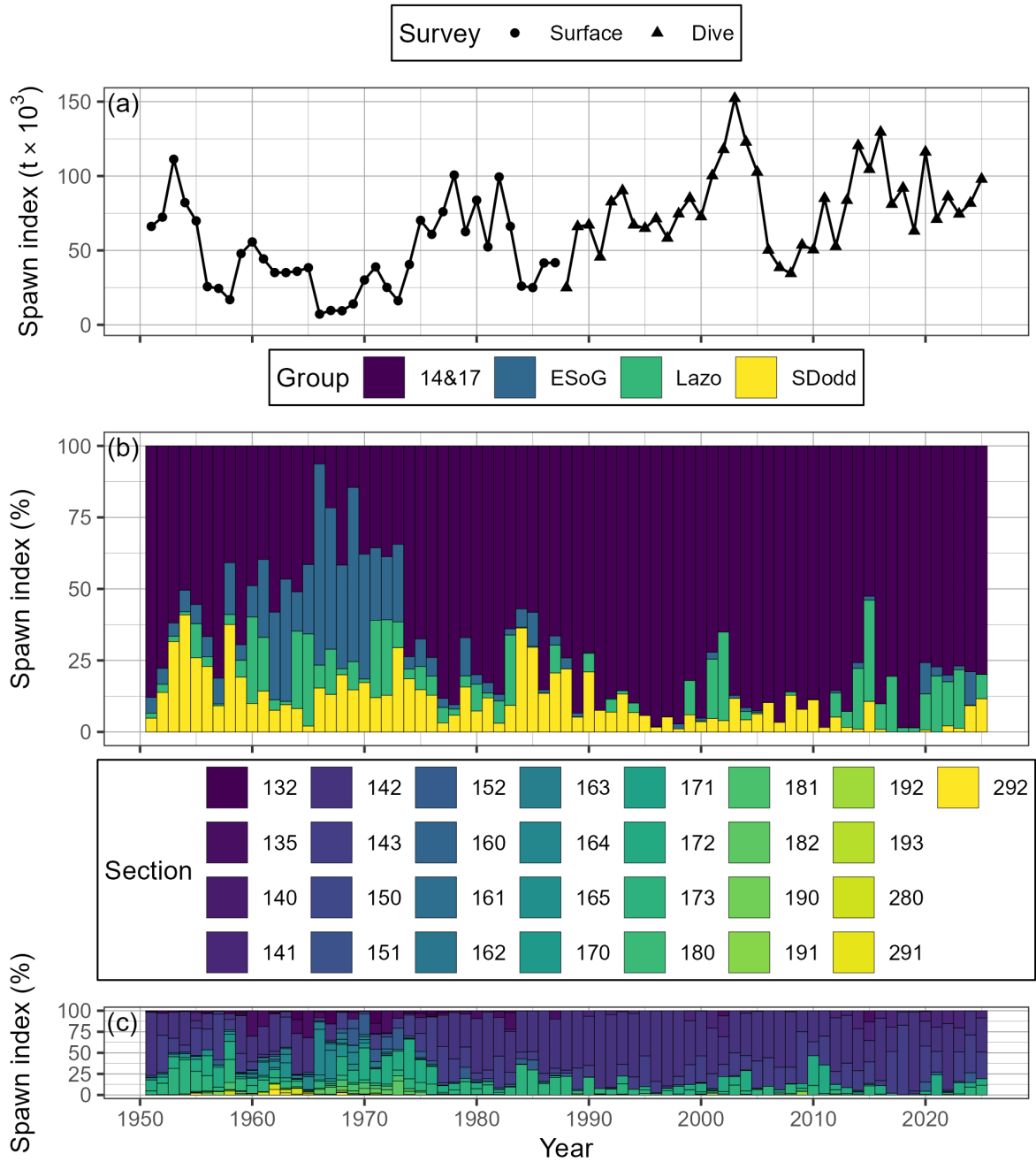


Figure 18. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR; panel a), as well as percent contributed by Group, and Section (b, & c, respectively). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). Note that spawn surveys in the dive survey period (1988 to 2025) are a combination of surface and dive surveys (Figures 15 and 16). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Legend: ‘14&17’ is Statistical Areas 14 and 17 (excluding Section 173); ‘ESoG’ is eastern Strait of Georgia; ‘Lazo’ is above Cape Lazo; and ‘SDodd’ is South of Dodd Narrows

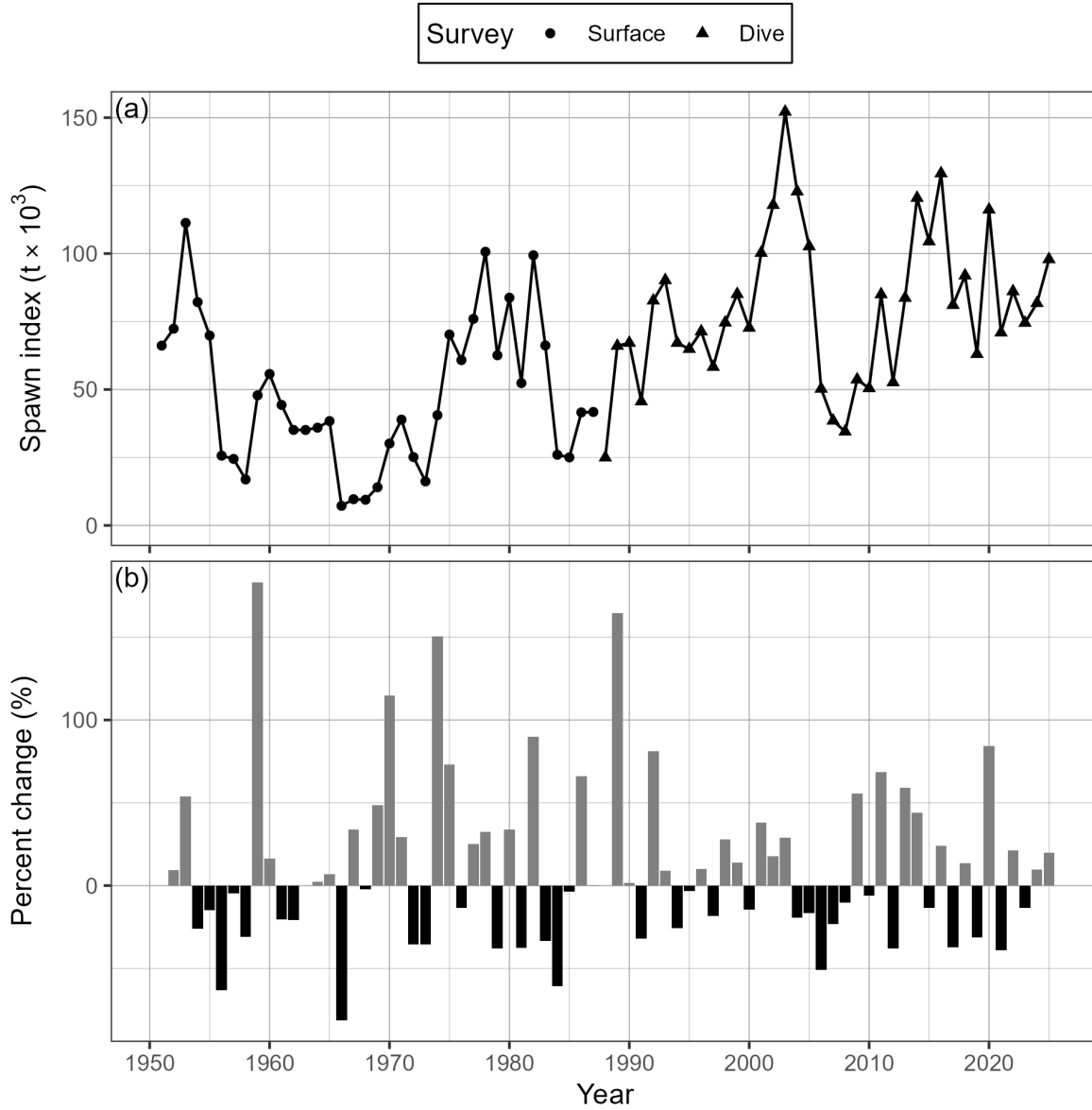


Figure 19. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR; panel a), and percent change (b). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the spawn index in year t . The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Note that spawn surveys in the dive survey period (1988 to 2025) are a combination of surface and dive surveys (Figures 15 and 16).

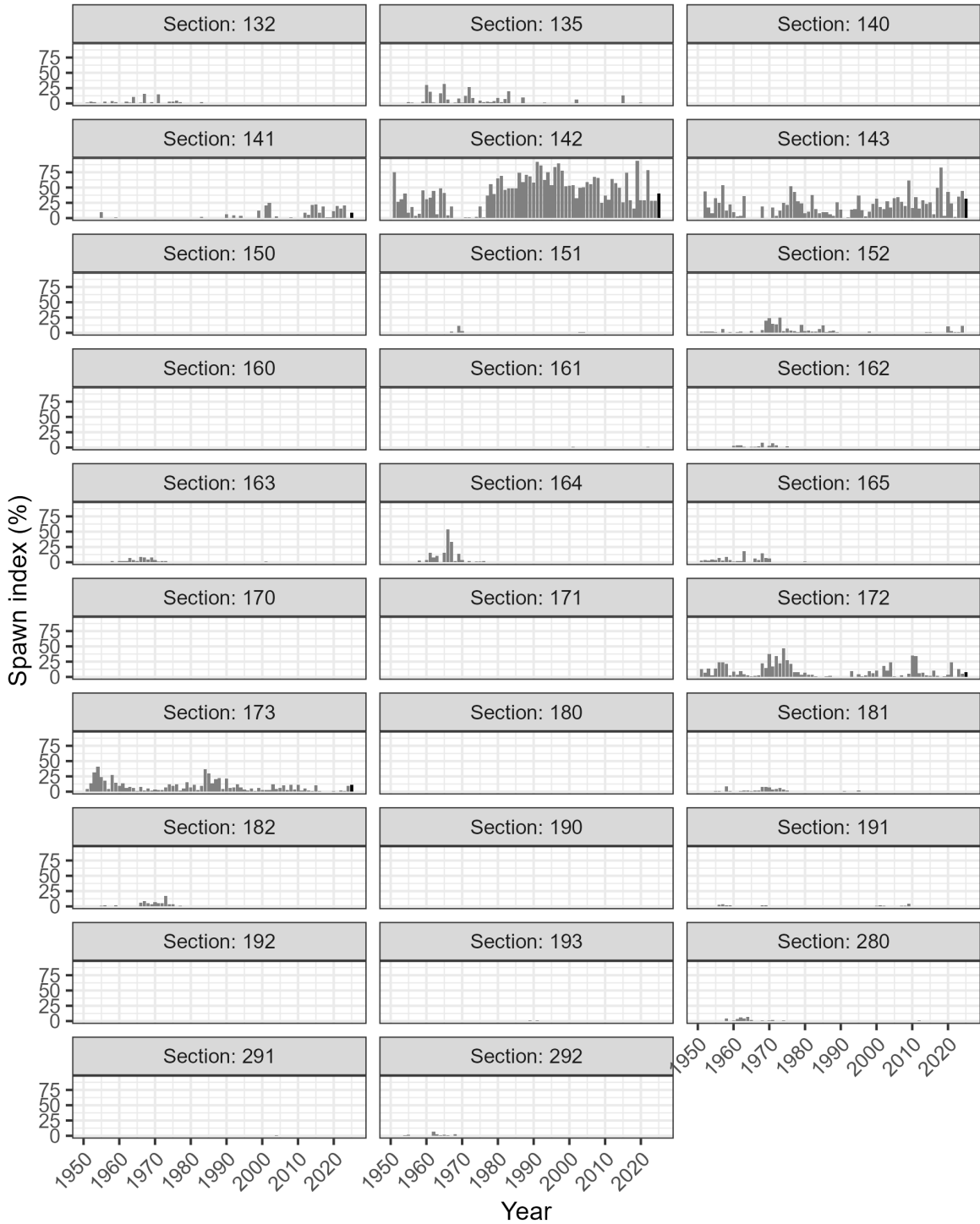


Figure 20. Time series of percent of spawn index by Section for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). The year 2025 has a darker bar to facilitate interpretation. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q .

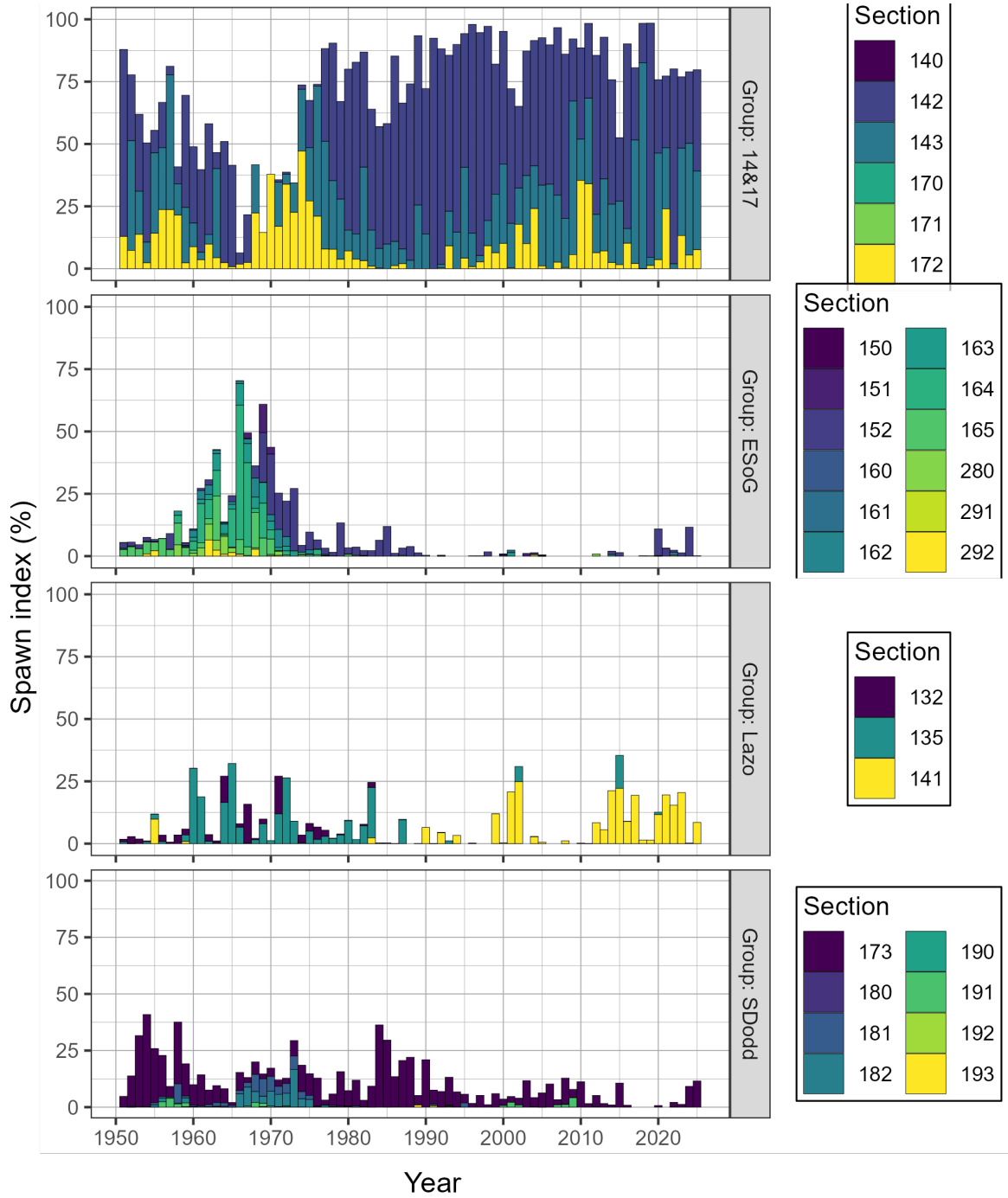


Figure 21. Time series of percent of spawn index by Group and Section for Pacific Herring from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Legend: ‘14&17’ is Statistical Areas 14 and 17 (excluding Section 173); ‘ESoG’ is eastern Strait of Georgia; ‘Lazo’ is above Cape Lazo; and ‘SDodd’ is South of Dodd Narrows

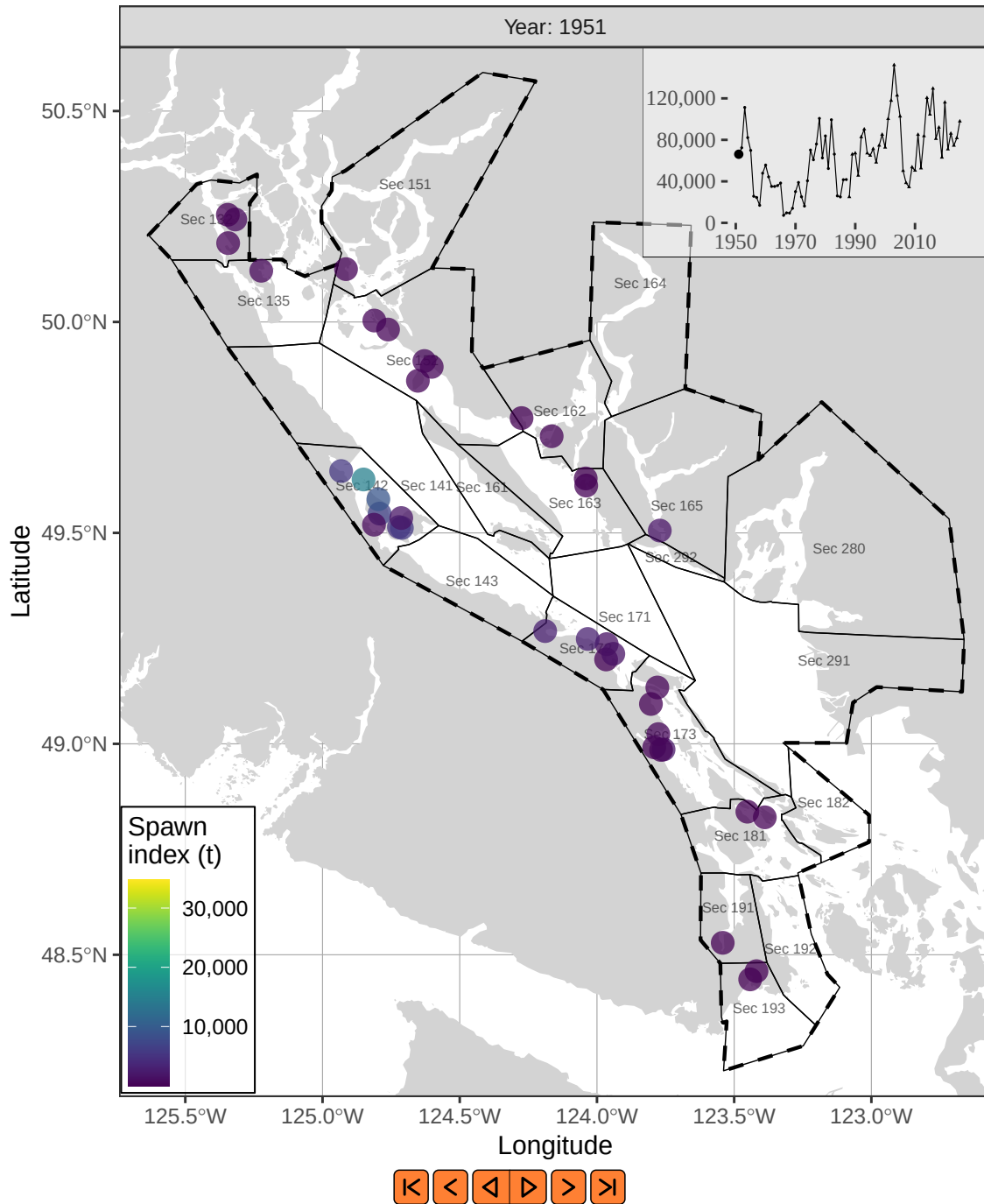


Figure 22. Animation of Pacific Herring spawn index in metric tonnes (t) by Location from 1951 to 2025 in the Strait of Georgia major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km). View the animation: [download the report](#), open with Adobe, enable Java, and click “play”.